

Venkata Sagar Maddela

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Education

Indian Institute of Information Technology - Sri city

Nov 2022 – Present

Bachelor of Science in Computer Science

- GPA: 8.20
- **Coursework:** Data Structures and Algorithms(C++), Object Oriented Programming, Database Management Systems (SQL), Operating Systems, Artificial Intelligence and Machine Learning (Python), Communication Networks and Cyber Security, Cloud Computing, Computer Vision(Python), Full-Stack Development (HTML, CSS, JS, Node, Express, MongoDB, React), High Performance Computing

Projects

PDFnquiry - RAG Powered PDF-Assistant

[github.com](#) 

- Developed an AI-powered PDF assistant enabling natural language queries over documents using **Retrieval-Augmented Generation** (RAG) with **Langchain** and **Ollama** models (**LLaMA3**, **Mistral**)
- Built a responsive frontend with React and Tailwind CSS; integrated with a **FastAPI** backend handling PDF uploads, vector indexing (**FAISS**), and question answering via local LLMs.
- Implemented semantic retrieval by chunking, embedding, and indexing PDF content for accurate and context-aware responses through a chat interface.
- **Tech Used:** React.js, Tailwind CSS, FastAPI, Python, FAISS, LangChain, Ollama, Docker Compose

NER-RoBERTa – Fine-Tuned Named Entity Recognition Model

[huggingface.co](#) 

- Developed a **RoBERTa**-based token classification model fine-tuned on custom annotated resume data to extract structured entities like personal info, education, work experience, skills, and more.
- Enabled **automated resume parsing** and **HR document understanding** with high precision using advanced token classification techniques.
- Leveraged **Huggingface Transformers** and **SpaCy** for training, inference, and preprocessing of diverse document formats (.pdf, .docx).
- Tech Used: Python, PyTorch, Huggingface Transformers, SpaCy, RoBERTa, Resume Parsing, Named Entity Recognition

Customer Segmentation using Gaussian Mixture Models

[GitHub](#) 

- Implemented an **unsupervised learning pipeline** combining **PCA for dimensionality reduction** and **GMM for probabilistic clustering**, enabling effective segmentation of customer data.
- Used **Bayesian Information Criterion (BIC)** for optimal cluster selection and **Silhouette Score** for evaluation. Visualized clusters in **2D and 3D**.
- **Tools Used:** Python, NumPy, Pandas, Scikit-learn, Matplotlib, StandardScaler, OneHotEncoder

Virtual Canvas – AI-Powered Hand Gesture Drawing App

[GitHub](#) 

- **Tools Used:** Python, OpenCV, MediaPipe, NumPy, Computer Vision

Certifications

Data structures and Algorithms specialization : Issued by University of California, San Diego (verified)

Publications

NER-RoBERTa: Fine-Tuned Named Entity Recognition Model [Publication Site](#) 

Skills

Languages: C, C++, Java, SQL, Python

Web Development : JavaScript, ReactJS, NodeJS, HTML, CSS, ExpressJS, MongoDB, TailwindCSS, FASTAPI

ML and Gen AI : Python, TensorFlow, Pytorch, Scikit-learn, OpenCV, MediaPipe, Langchain , Local LLM, Mistral, Transformers, FAISS embeddings, Named-Entity-Recognition, RoBERTa, SpaCy.

Research : High Performance Computing, Cryptography.

Tools : AWS, Vercel, Git, Github, Linux, Docker, OpenMP, CUDA, MPI